

VIRO - AI

Viral Insight & Rapid Optimization Analytics Intelligence

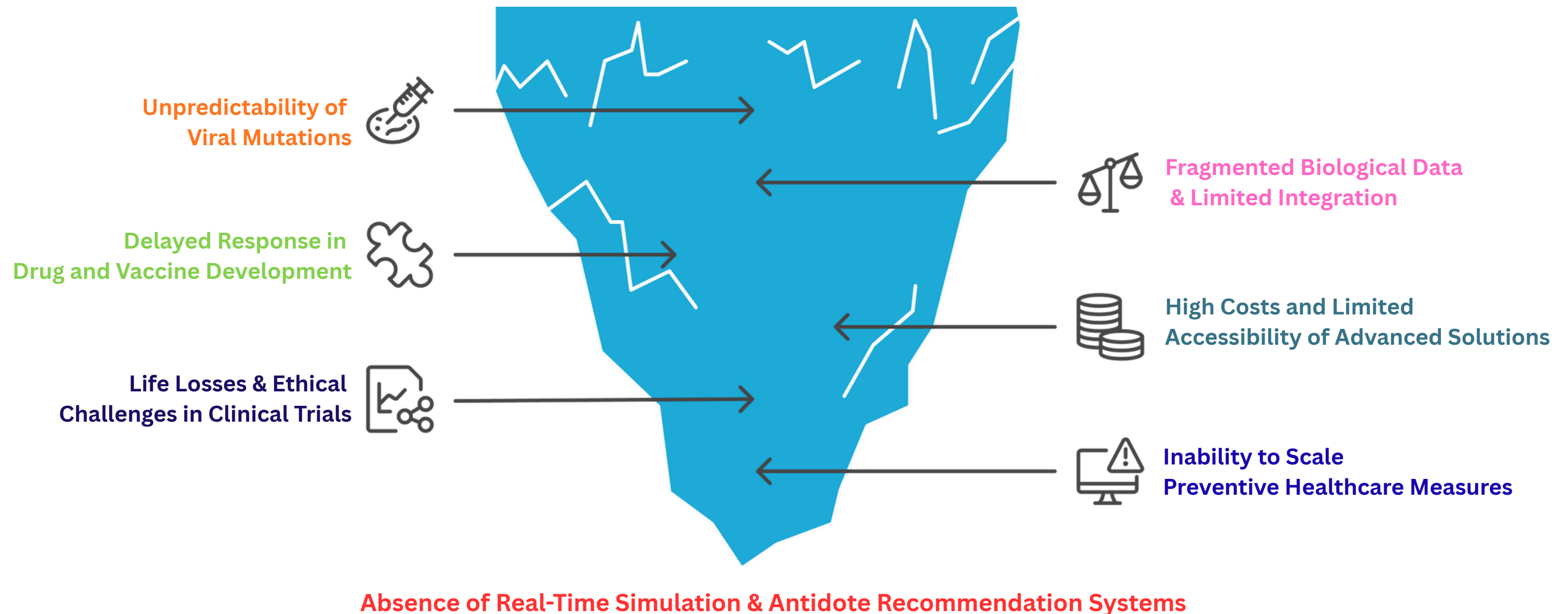
Team Cubic Vision



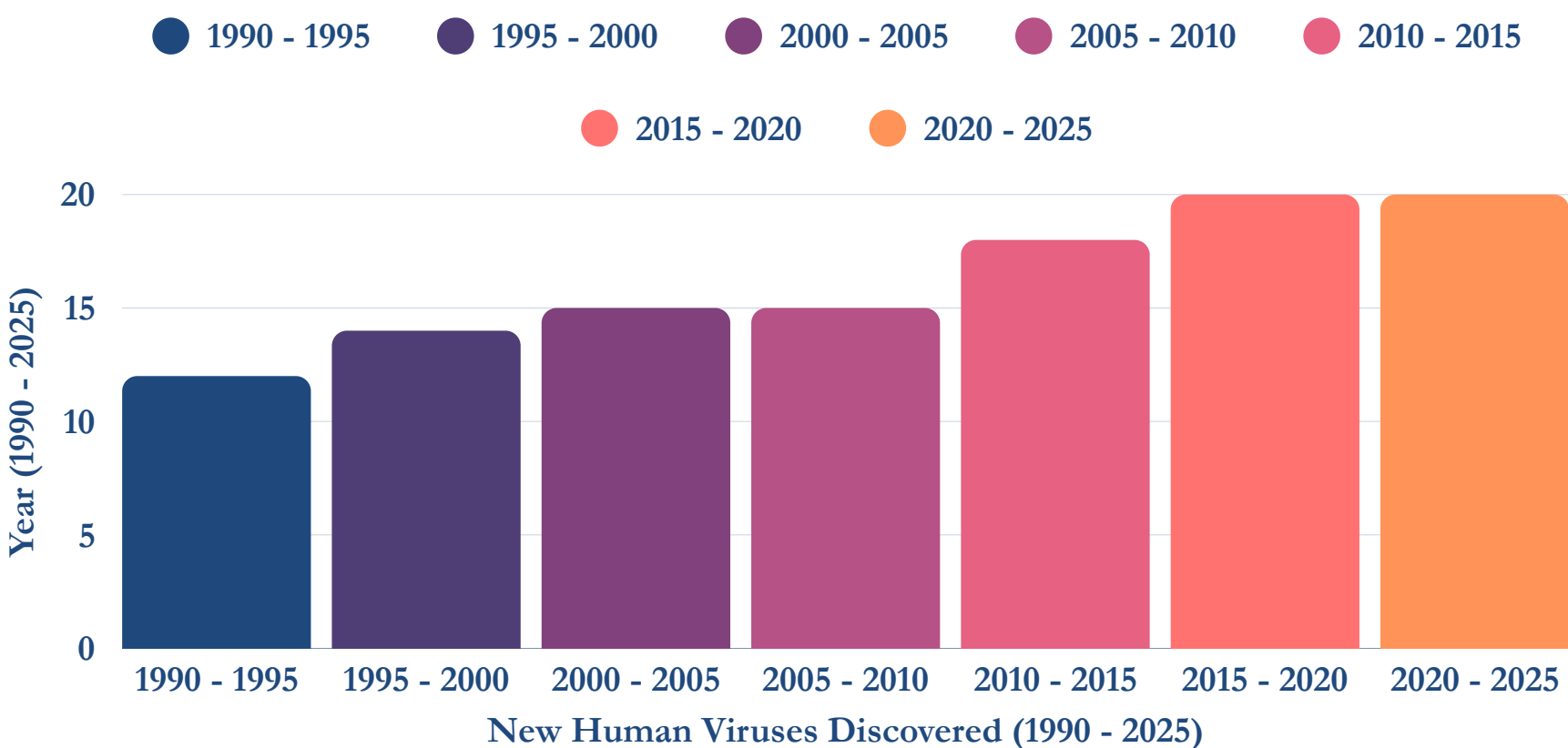


PROBLEM STATEMENT

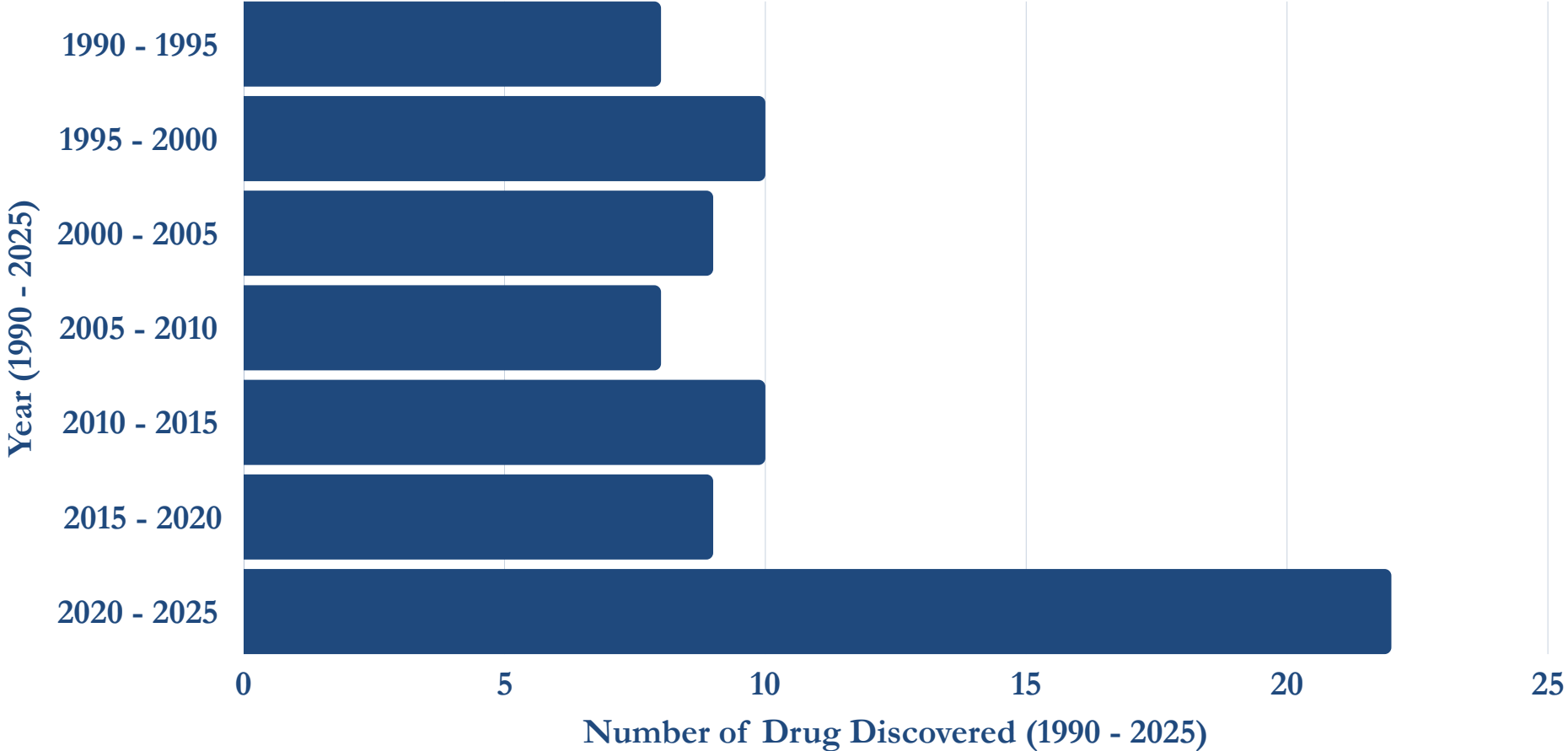
The rapid mutation of viruses impedes timely vaccine and drug development, resulting in delayed outbreak response and increased mortality. Current systems lack predictive precision and integration between genomic analytics and therapeutic modeling. VIRO–AI addresses this gap by leveraging artificial intelligence to forecast viral mutations, simulate molecular interactions, and accelerate the discovery of effective antivirals—enabling proactive, data-driven responses to evolving infectious diseases.



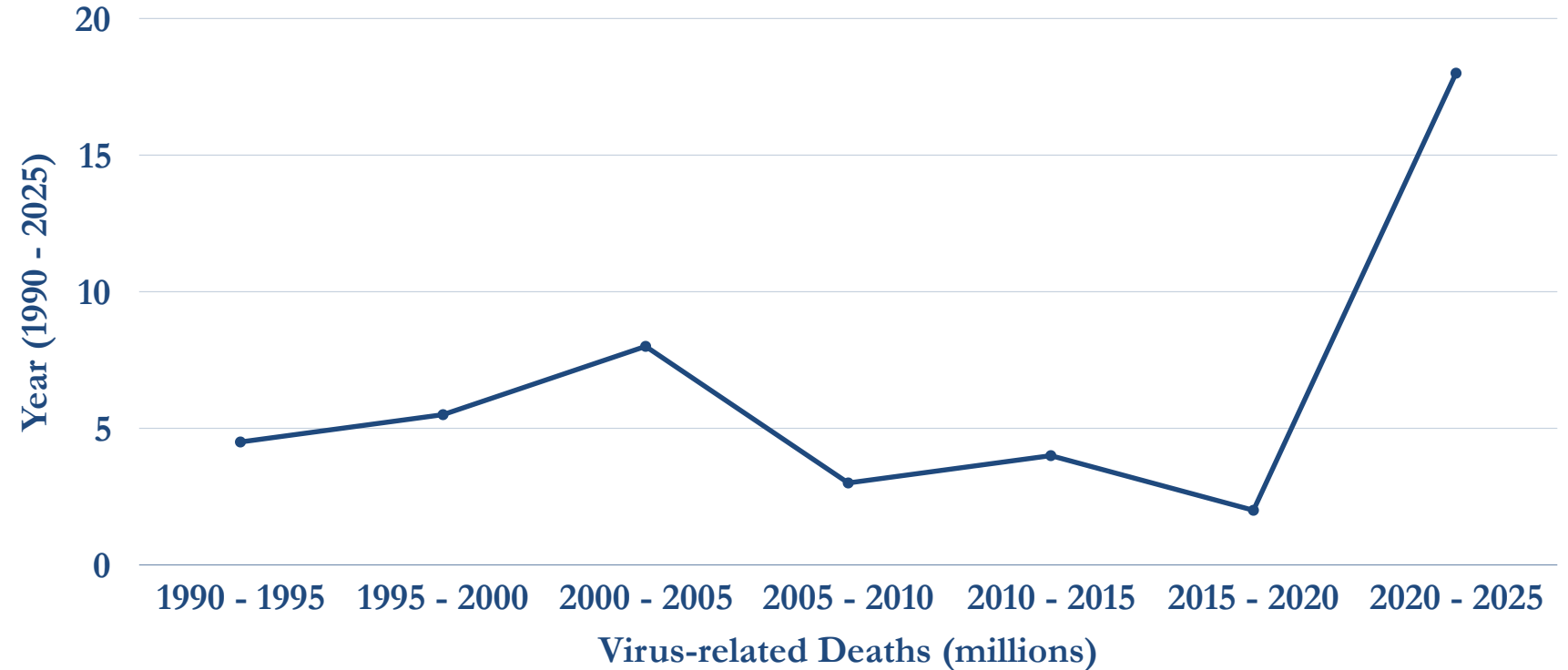
New Human Viruses Discovered (1990 - 2025)



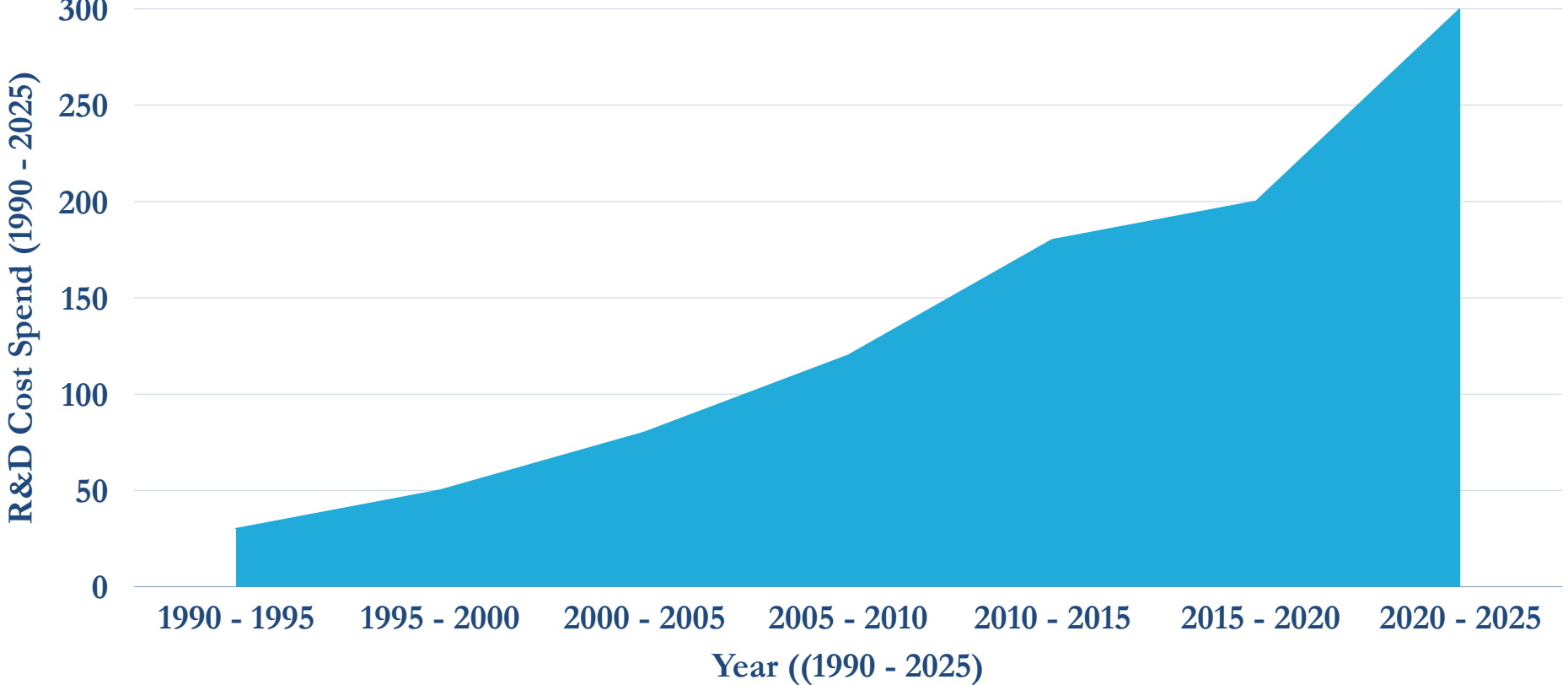
Antidotes Discovered (1990 - 2025)



Virus-related Deaths (millions)



Reserach & Development Cost in Drug Discovery(1990 - 2025)



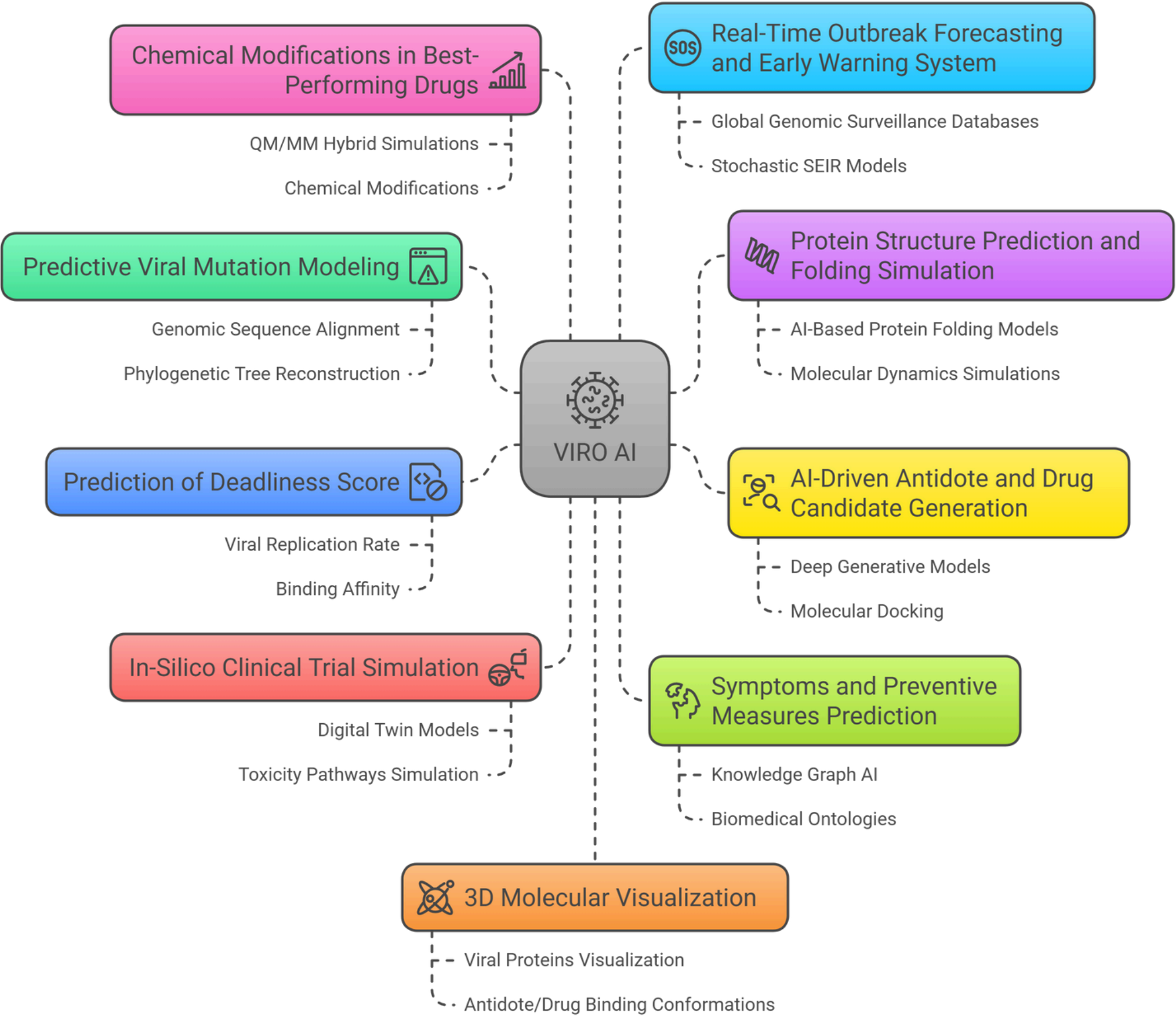


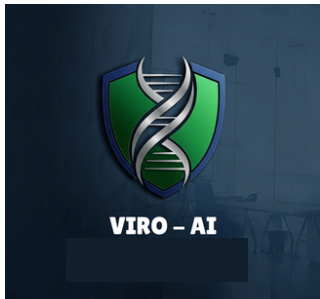
SOLUTION

VIRO-AI is an advanced AI-driven bioinformatics platform designed to predict viral mutations, simulate protein-ligand interactions, and accelerate antiviral discovery. It leverages artificial intelligence, molecular modeling, and evolutionary computation to analyze viral behavior at the genomic and structural levels, offering a transformative approach to predictive virology and therapeutic innovation.

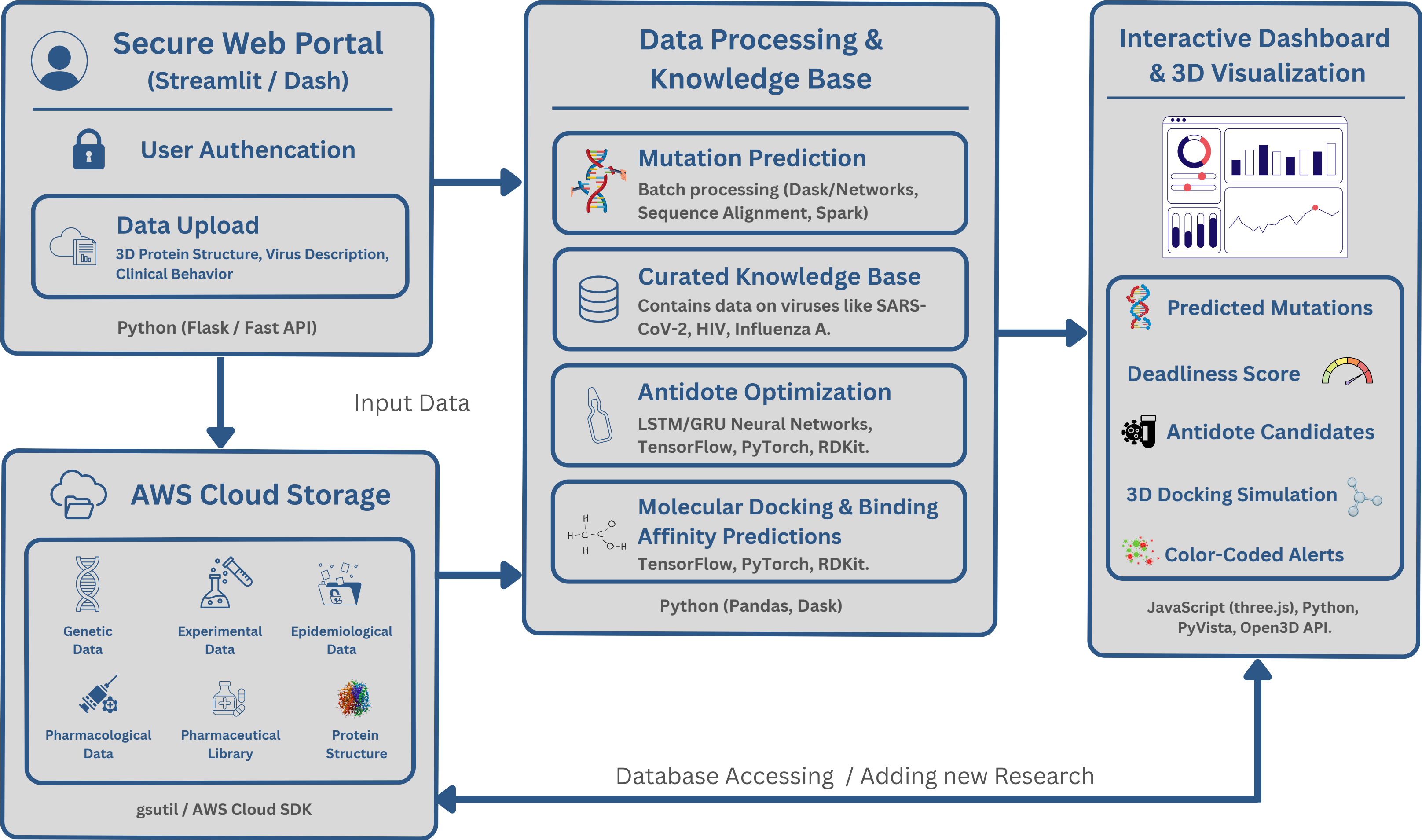
The platform integrates genomic analysis, evolutionary modeling, and molecular dynamics simulations to forecast mutation pathways and evaluate their structural and functional impacts. By applying deep learning and generative algorithms, **VIRO-AI** designs novel drug and antibody candidates while computing pathogenicity scores, immune evasion potential, and outbreak risks with high precision.

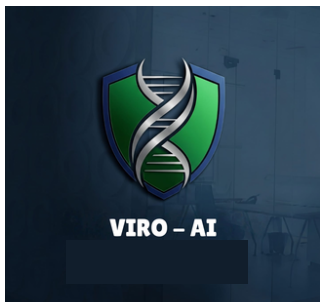
Through its cloud-based architecture, **VIRO-AI** delivers real-time insights and interactive dashboards that support data-driven decision-making for researchers, pharmaceutical industries, and public health authorities. This enables proactive outbreak management, reduces drug development time, and strengthens global pandemic preparedness through predictive, adaptive, and intelligent virological analytics.





SYSTEM ARCHITECTURE





INNOVATION & UNIQUENESS

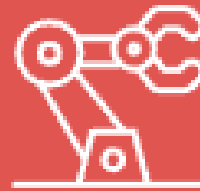
Mutation Forecasting

Predicts viral mutations for better preparedness.



Drug Generation

AI generates drugs and antidotes quickly.



Symptom Prediction

Predicts symptoms and preventive measures.



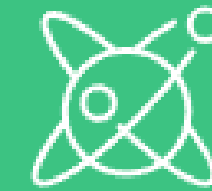
Molecular Visualization

Visualizes virus-drug interaction in 3D.



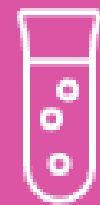
Bioinformatics Infrastructure

Scalable cloud-based bioinformatics infrastructure.



Deadliness Scoring

Scores deadliness and pathogenicity of viruses.



Chemical Modification

Modifies lead compounds for improved efficacy.



Trial Simulator

Simulates clinical trials in-silico for faster results.



Outbreak Forecasting

Forecasts outbreaks in real-time for rapid response.

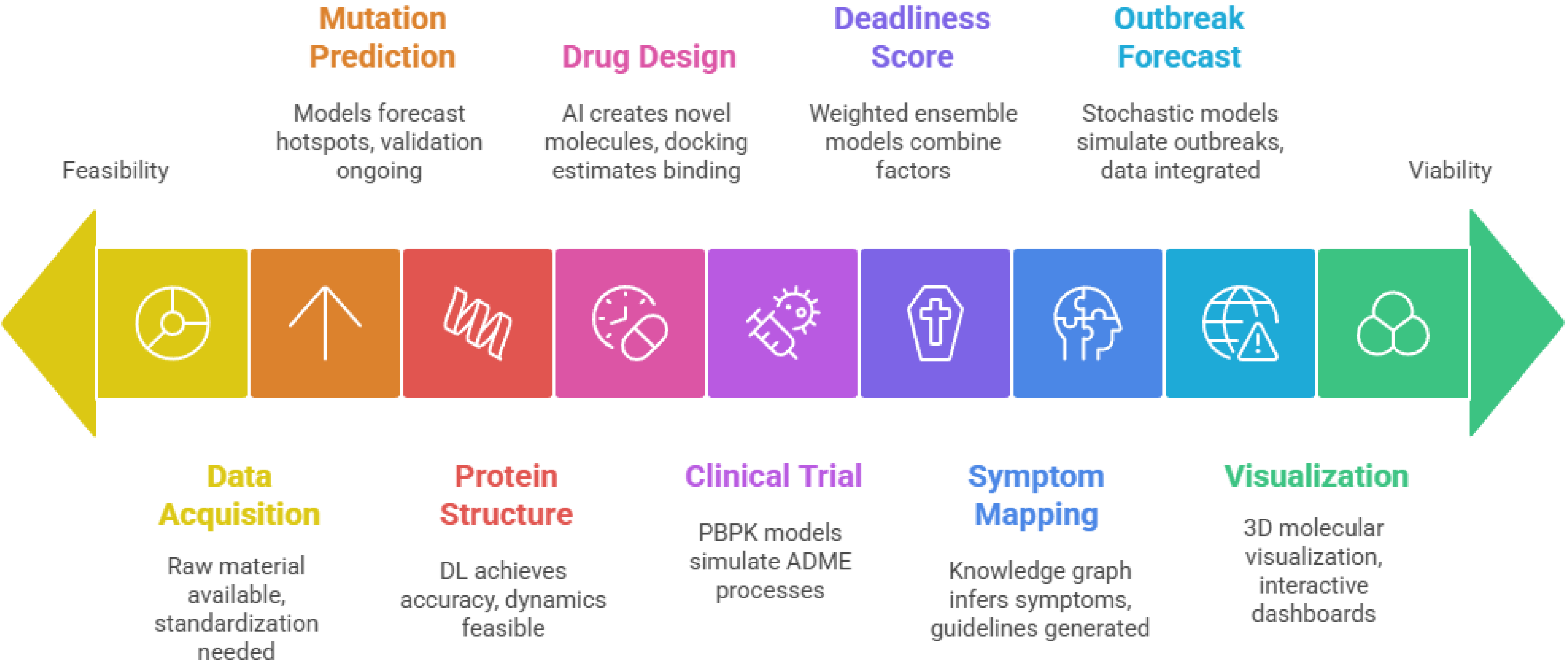


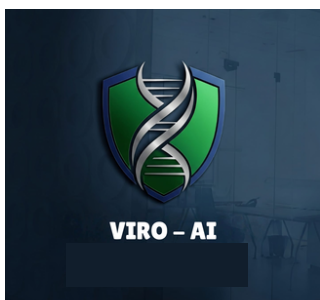
Translational Pipeline

End-to-end translational pipeline for drug development.



FEASIBILITY & VIABILITY





IMPACT & BENEFITS

Drug Discovery

Accelerates antiviral drug discovery using deep generative chemistry and virtual screening.

Clinical Trials

Reduces mortality and ethical risks in clinical trials through in-silico simulations.

Viral Evolution

Predicts viral evolution using genetic algorithms and evolutionary simulations.

Pandemic Preparedness

Improves pandemic preparedness and outbreak forecasting using epidemiological models.

Systemic Impact

Predicts organ-level and systemic impact using systems biology models.

Drug Optimization

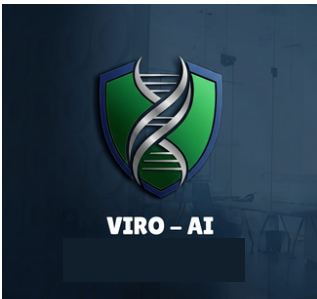
Enhances drug optimization and chemical refinement using QM/MM simulations.

Democratization

Democratizes computational virology with cloud-native, GPU-accelerated architecture.

Visualization

Enhances visualization and interpretability with interactive 3D molecular visualization.



BUSINESS IDEOLOGY

Characteristic	Phase 1	Phase 2	Phase 3	Phase 4
 Target Audience	Academic/research labs, ICMR, startups	Pharma companies, top pharma, CROs	National health agencies, state labs, hospitals	Central government, national health missions, PPPs
 System Accuracy	Mutation prediction, protein folding, docking	Clinical trial simulation, drug-likeness, side-effect prediction	Outbreak forecasting, mutation-to-symptom, deadliness score	Multi-omics integration, vaccine prediction, outbreak prevention
 Statistical Data	Viral genomes processed, protein models, docking simulations	Clinical trial simulations, drug molecules screened, candidate reduction	Genomes processed, samples handled, outbreak forecasts	Institutions covered, genomes processed, pandemic impact reduction
 Revenue Potential	Pilot contracts with labs	SaaS/API licensing for pharma	SaaS subscription for ICMR, hospital partnerships	National licensing, pharma enterprise deals

The background features a light blue gradient with a series of concentric circles emanating from the center. Overlaid on these are numerous thin, curved lines in shades of grey and gold, which sweep across the frame from the left and right sides, creating a sense of dynamic movement and flow.

**VIRO-AI IS NOT JUST A TOOL IT'S A PARADIGM
SHIFT IN OUR FIGHT AGAINST INFECTIOUS
DISEASES.**

JOIN US IN BUILDING A HEALTHIER, SAFER FUTURE.



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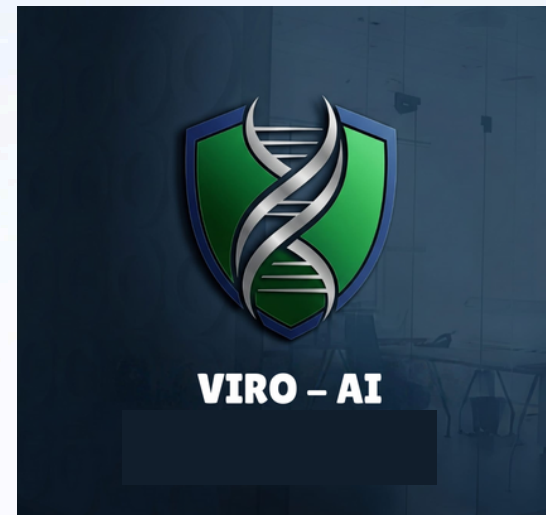


Git Hub Profile

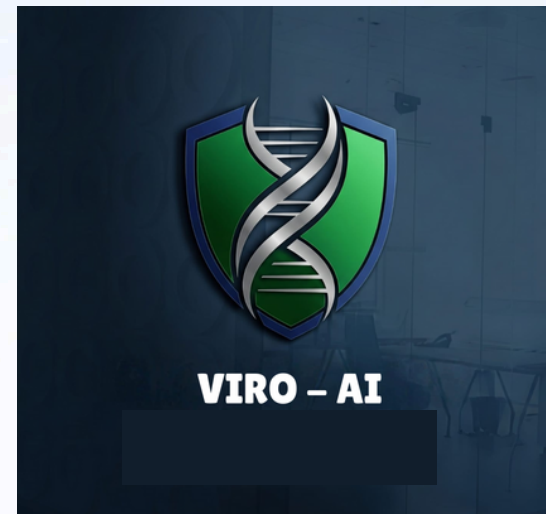


SCAN ME





ANY QUESTIONS / SUGGESTIONS ??



THANK YOU